CAMBRIDGE
International Education

# Cambridge International AS & A Level

**MATHEMATICS** **9709/41**

Paper 4 Mechanics **October/November 2025**

MARK SCHEME

Maximum Mark: 50

**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.



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## Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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**Mathematics-Specific Marking Principles**

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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**Annotations guidance for centres**

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

**Annotations**

| Annotation | Meaning |
|---|---|
|  | More information required |
|  | Accuracy mark awarded zero |
|  | Accuracy mark awarded one |
|  | Independent accuracy mark awarded zero |
|  | Independent accuracy mark awarded one |
|  | Independent accuracy mark awarded two |
|  | Benefit of the doubt |
|  | Blank Page |
|  | Incorrect |
| Dep | Used to indicate DM0 or DM1 |

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| Annotation | Meaning |
|---|---|
| DM1 | Dependent on the previous M1 mark(s) |
| FT | Follow through |
| ~~~ | Indicate working that is right or wrong |
| Highlighter | Highlight a key point in the working |
| ISW | Ignore subsequent work |
| J | Judgement |
| JU | Judgement |
| M0 | Method mark awarded zero |
| M1 | Method mark awarded one |
| M2 | Method mark awarded two |
| MR | Misread |
| O | Omission or Other solution |
| Off-page comment | Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to. |
| On-page comment | Allows comments to be entered in speech bubbles on the candidate response. |
| PE | Judgment made by the PE |
| Pre | Premature approximation |
| SC | Special case |
| SEEN | Indicates that work/page has been seen |

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| Annotation | Meaning |
|---|---|
| SF | Error in number of significant figures |
| ✓ | Correct |
| TE | Transcription error |
| XP | Correct answer from incorrect working |

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## Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

### Types of mark

**M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

**A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

**B** Mark for a correct result or statement independent of method marks.

**DM** or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

**FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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**Abbreviations**

| | |
|---|---|
| AEF/OE | Any Equivalent Form (of answer is equally acceptable) / Or Equivalent |
| AG | Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid) |
| CAO | Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed) |
| CWO | Correct Working Only |
| ISW | Ignore Subsequent Working |
| SOI | Seen Or Implied |
| SC | Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance) |
| WWW | Without Wrong Working |
| AWRT | Answer Which Rounds To |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 1(a) | $D - 350 = 900 \times 0.25$ | **M1** | Attempt at N2L – correct number of terms. Allow sign errors but must be dimensionally correct. |
| | $D = 575$ N | **A1** | |
| | | **2** | |
| 1(b) | $P = 575 \times 15$ | **M1** | Use of $P = D \times v$ with their $D$ from **1(a)** and $v$ = 15. |
| | $P = 8625$ W or 8.625 kW | **A1** | Allow 8625 without units,<br>but 8.625 must have kW. |
| | | **2** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 2(a) | $3\times4=3\times(-1)+5\times v$<br><br>**or** $3\times4=3\times1+5v$ | **M1** | Attempt at conservation of linear momentum; three non-zero terms – allow sign errors. **M1** only if using weight rather than mass. |
| | $v=3\,\text{m s}^{-1}$ | **A1** | |
| | $v=1.8\,\text{m s}^{-1}$ | **A1** | If **A0 A0 SC B1** for both $-3$ and $-1.8$. |
| | | **3** | |
| 2(b) | KE = $\frac{1}{2}\times3\times4^2\ (=24)$ **or** KE = $\frac{1}{2}\times3\times1^2+\frac{1}{2}\times5\times1.8^2\ (=9.6)$<br><br>**or** KE = $\frac{1}{2}\times3\times1^2+\frac{1}{2}\times5\times3^2\ (=24)$ | **M1** | Attempt at either the total kinetic energy before or after the collision.<br>The total KE (before and after) could be embedded in a calculation for the KE loss.<br>**M1** be awarded just on sight of 24, 19.6 or 14.4. |
| | $\lambda=14.4$ ONLY | **A1** | Allow –14.4.<br>Must have discarded $\lambda=0$ if mentioned. |
| | | **2** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| | **NO MISREADS IN THIS QUESTION** | | |
| 3 | For attempt to resolve in any direction to form an equation or expression | ***M1** | Correct number of terms; allow sign errors and sin/cos mix ONLY. |
| | Vertically: $(Y=)\ \pm(45+28\sin 35-72\sin 50-35\sin 60)$ | **A1** | $\pm 24.405948\ldots$ |
| | Horizontally: $(X=)\ \pm(28\cos 35+72\cos 50-35\cos 60)$ | **A1** | $\pm 51.716965\ldots$ |
| | $R=\sqrt{(their\,24.4059...)^2+(their\,51.7169...)^2}$ | **DM1** | Attempt to solve for $R$ from equations with the correct number of relevant terms (only sign errors and sin/cos mix allowed). |
| | $\theta=\tan^{-1}\left(\frac{their\,24.4059...}{their\,51.7169...}\right)\left[=\tan^{-1}0.4719...\right]$<br><br>OR $\alpha=\tan^{-1}\left(\frac{their\,51.7169...}{their\,24.4059...}\right)\left[=\tan^{-1}2.1190...\right]$ | **DM1** | Attempt to solve for $\theta$ or $\alpha$ from equations with the correct number of relevant terms (only sign errors and sin/cos mix allowed). |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 3 | $R = 57.2$ N<br><br>Direction is $25.3^\circ$ below the positive $x$-direction<br>or bearing $115^\circ$ | **A1** | Both correct.<br><br>Giving an angle only is insufficient. Direction may be seen on a diagram, with minimum of arrow on resultant OE.<br><br>May give direction relative to one of the given forces e.g. $24.7^\circ$ anticlockwise from the 72 N force.<br><br>Arrows on both components only is **A0** as it doesn't show the direction of the resultant.<br><br>However the direction is stated, it must be able to be drawn uniquely. |
| | | **6** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 4 | PE lost by $A$: $\pm mg \times 0.75$ | **B1** | |
| | PE gained by $B$: $\pm 3g \times 0.75 \times 0.6 \left[ = \pm 13.5 \right]$ | **B1** | Allow $\sin 37$ or better for 0.6.<br>$\sin^{-1} 0.6 = 36.86989765$ . |
| | KE gained by $A$ **and** $B$: $\pm\left( \frac{1}{2} \times 3 \times 2^2 + \frac{1}{2} \times m \times 2^2 \right) \left[ = \pm(6 + 2m) \right]$ | **B1** | |
| | Work done against resistance = $\pm \frac{10}{3} \times 0.75 \left[ = \pm 2.5 \right]$ | **B1** | |
| | $\frac{1}{2} \times 3 \times 2^2 + \frac{1}{2} \times m \times 2^2 = 0.75mg - 13.5 - 2.5$ | **M1** | Attempt at work-energy equation. Equivalent of 5 terms, dimensionally correct, allow sign errors, PE gained by $B$ must include a component of $3g$ . |
| | $m = 4$ | **A1** | |
| | **Special case for use of N2L max 3 marks** | | |
| | $2^2 = 0^2 + 2 \times a \times 0.75 \Rightarrow a = \frac{8}{3}$ | **B1** | |
| | $mg - T = ma$ **and** $T - 3g \times 0.6 - \frac{10}{3} = 3a$<br><br>OR $mg - 3g \times 0.6 - \frac{10}{3} = (m + 3)a$ | **M1** | Correct number of dimensionally correct terms, allow sign errors, allow sin/cos mix.<br>Allow $\sin 37$ or better for 0.6.<br>$\sin^{-1} 0.6 = 36.86989765$ . |
| | $m = 4$ | **A1** | |
| | | **6** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 5 | Attempt at resolving parallel or perpendicular to the plane to form an equation | ***M1** | Correct number of terms, allow sign errors, allow sin/cos mix, allow $g$ missing on the weight term only (so **not** missing on the $8mg$ term).<br><br>Forces that need resolving should be resolved, forces that do not need resolving should not be resolved. |
| | $8mg\cos\theta + mg\sin\theta = F$ | **A1** | Allow with their possibly incorrect $F$ . |
| | $R = mg\cos\theta + 8mg\sin\theta$ | **A1** | |
| | Use of $F = 0.5R$ to get an equation in $\theta$ [ $m$ and $g$ ] | **DM1** | Where $R$ is a linear combination of a weight (or mass) component and an $8mg$ component. |
| | $8mg\cos\theta + mg\sin\theta = 0.5\left(mg\cos\theta + 8mg\sin\theta\right)$<br><br>$\Rightarrow \tan\theta = 2.5$ | **DM1** | Re-arranging to $\tan\theta = k$ where $k \neq 0$ (or equivalent method to reduce to a single trig. term). – dependent on both previous **M1** marks. |
| | $\theta = 68.2$ | **A1** | |
| | | **6** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 6 | Acceleration of $A$ down the plane is $g\times0.2\left[=2\right]$ | **B1** | Allow $-g\times0.2$.<br>Allow $\pm g\times\sin12$ or better.<br>Allow $2.5a=2.5g\times0.2$. |
| | Speed of $A$ before collision is $g\times0.2\times2\left[=4\right]$ | ***M1** | Use of $v=u+at$ with $u$ = 0, $t$ = 2 and their $a$ which must be equivalent to either $\pm g\times\sin12$ or $\pm g\times\cos12$ ONLY. |
| | $3\times3.5+2.5\times\left(-4\right)=3\times\left(-0.5\right)+2.5\times u$ | **DM1** | Attempt at conservation of linear momentum; four non-zero terms – allow sign errors. **M1A0** only if using weight rather than mass. |
| | $u=0.8$ | **A1** | Correct speed (or velocity) of $A$ after impact. |
| | $0=0.8^2+2\times\left(-g\times\frac{1}{5}\right)\times s$ | **DM1** | Use of $v^2=u^2+2as$ (or other complete method to find $s$) with $v=0$, their $u$ and their $-a$ – condone sign errors. **Dependent on both previous M1 marks.** |
| | $s=0.16$ m | **A1** | CWO. |
| | | **6** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 7(a) | $v=\frac{1}{3}\left(2t^2-11t+12\right)\Rightarrow\frac{\mathrm{d}v}{\mathrm{d}t}=\frac{1}{3}\left(4t-11\right)$ | **M1** | Expand to a 3-term quadratic equivalent and attempt to differentiate (oe e.g. product rule), decrease power by 1 and a change in coefficient in at least one term; allow un-simplified<br>$a=\frac{v}{t}$ is **M0**. |
| | $\frac{1}{3}\left(4\times2-11\right)=-1\text{m s}^{-2}$ | **A1** | Must be negative. |
| | | **2** | |
| 7(b) | Attempt to integrate | ***M1** | Expand to a 3-term quadratic equivalent and attempt to integrate, increase power by 1 and a change in coefficient on the same term.<br>$s = vt$ is **M0.** |
| | $s=\frac{1}{3}\left(\frac{2t^3}{3}-\frac{11t^2}{2}+12t\right)\left(+c\right)$ | **A1** | Allow un-simplified. |
| | [$P$ changes direction when] $t = 1.5$ | ***B1** | $t = 1.5$ SOI. |
| | Attempt to evaluate their $\frac{1}{3}\left(\frac{2t^3}{3}-\frac{11t^2}{2}+12t\right)$ for $t = 0$ to $t = 1.5$ **and** $t = 1.5$ to $t = 3$ | **DM1** | Using the correct limits correctly. |
| | Total distance is $\left[2.625+\left\|-1.125\right\|=\right]$ 3.75 m | **A1** | If no integration seen, then **B1** for 3.75 (so 2 marks max.). |
| | | **5** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 7(c) | Attempt to evaluate their $s$ for $t = 0$ to $t = 4$ | **M1** | OE e.g. compare $\int_0^{1.5} v\,\mathrm{d}t$ with $\int_{1.5}^{4} v\,\mathrm{d}t$<br>**or** setting their three-term cubic expression in $t$ (with no constant term) equal to zero and attempting to solve or attempt discriminant. |
| | $\int_0^4 v\,\mathrm{d}t = \frac{8}{9} > 0$ so, no $P$ does not return to $O$ | **A1** | OE e.g. obtaining $4t^2 - 33t + 72 = 0$ and showing that this has no real roots (and hence $P$ does not return to $O$).<br>$4t^2 - 33t + 72 = 0$ gives<br>$b^2 - 4ac = (-33)^2 - 4 \times 4 \times 72 = -63$.<br>$\frac{2}{9}t^2 - \frac{11}{6}t + 4 = 0$ gives<br>$b^2 - 4ac = \left(-\frac{11}{6}\right)^2 - 4 \times \frac{2}{9} \times 4 = -\frac{7}{36}$. |
| | | **2** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 8(a) | $T\cos 15 = 0.5\times 1.2$ | **M1** | N2L for $B$ – correct number of terms, allow sin/cos mix ONLY. |
| | $T = 0.621\,\text{N}$ | **A1** | Allow $\dfrac{3\sqrt{6}-3\sqrt{2}}{5}$. |
| | | **2** | |
| 8(b) | Attempt to resolve vertically for $A$ to form an equation | ***M1** | Correct number of terms, allow sign errors and sin/cos mix, allow their (possibly incorrect) $T$ from **8(a).** |
| | $6\sin 20 + R_A = 2g + T\sin 15$ | **A1** | $R_A = 18.10864\ldots$ |
| | Attempt at N2L for $A$ | ***M1** | Correct number of terms, allow sign errors and sin/cos mix, allow their (possibly incorrect) $T$ from **8(a).** |
| | $6\cos 20 - T\cos 15 - F = 2\times 1.2$ | **A1** | $F = 2.63815\ldots$ |
| | Use of $F = \mu R$ to get an equation in $\mu$ only | **DM1** | **Dependent on both previous M1 marks.** |
| | $\mu = 0.146$ | **A1** | $\mu = 0.1456848\ldots$ |
| | | **6** | |

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